



YEAR 11 MATHEMATICS EXTENSION 1

Task 2

21 June 2024

General instructions

- Working time – 50 min.
- Write using a blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NSB approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **All 4** questions.

Class (please tick)

- ☐ 11 MM3.A1 – Ms Lee
- ☐ 11 MM3.B1 – Mr Berry
- ☐ 11 MM3.C1 – Mr Umakanthan
- ☐ 11 MM3.D1 – Dr Vranesevic
- ☐ 11 MM3.E1 – Ms Moss
- ☐ 11 MM3.F1 – Mr Ireland
- ☐ 11 MM3.G1 – Ms Cai / Mr Lott

STUDENT NUMBER

Marker’s use only.

QUESTION	1	2	3	4	Total	%
MARKS	$\overline{9}$	$\overline{11}$	$\overline{9}$	$\overline{16}$	$\overline{45}$	$\overline{100}$

Question 1. (9 Marks) (Use writing booklet 1)

Consider the real numbers a and b such that the equation in x ,

$$-|x - 1| - |x - 4| = ax + b \text{ *1. has a solution}$$

- (a) Set the left side of equation *1 as $y = -|x - 1| - |x - 4|$
This can be represented without using the absolute value signs in the following way.
- (i) If $x < \mathbf{A}$, then $y = \mathbf{B}x + \mathbf{C}$. Find the numbers $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ 1
 - (ii) If $\mathbf{A} \leq x < \mathbf{D}$, then $y = \mathbf{E}$. Find the numbers $\mathbf{D}$ and $\mathbf{E}$. 1
 - (iii) If $x \geq \mathbf{D}$, then $y = \mathbf{F}x + \mathbf{G}$. Find the numbers $\mathbf{F}$ and $\mathbf{G}$. 1
 - (iv) Sketch the graph of $y = -|x - 1| - |x - 4|$ on the grid provided in Answer Booklet 1. 2
- (b) Next, let us consider the common point(s) of the graphs of $y = -|x - 1| - |x - 4|$ and the straight line $y = ax + b$, on the xy -plane.
- (i) If $a = 1$, then the range of the values of b such that equation *1 has **one or more** solutions is $b \leq \mathbf{H}$. Find the value of $\mathbf{H}$. 2
 - (ii) If $b = -6$, then the range of the values of a such that equation *1 has **two distinct** solutions is $\mathbf{J} < a < \mathbf{L}$. Find the values of $\mathbf{J}$ and $\mathbf{L}$. 2

Question 2. (11 Marks) (Use writing booklet 2)

There are **2 identical black marbles** and **4 identical white marbles** arranged in a row.

- (i) How many different arrangements are possible? 1
- (ii) How many different arrangements of **just 4 of these** marbles are possible? 2
- (iii) Find the number of different words that can be formed by using all the letters in the word the **F A C E T I O U S** so that the vowels are in the same order. 3
- (iv) (i) If repetitions are **not** allowed, how many numbers **less than 4000** can be made from the digits **1, 2, 3, 4, 5, 6, 7**. 2
- (ii) How many of these numbers are even? 3

Question 3. (9 Marks) (Use writing booklet 3)

- (a) Find the independent term in the expansion of $\left(2x^2 - \frac{1}{x^3}\right)^{15}$ 2
- (b) Find n , given that the 2nd and 3rd coefficients in the expansion of $(2 + 3x)^n$ in ascending powers of x are in the ratio 4 : 45 3
- (c) Use the identity $1 + x + x^2 + \dots + x^n \equiv \frac{1 - x^{n+1}}{1 - x}$, where $x \neq 1$, to show that 4
- $1 - 2 + 3 - 4 + \dots = (-1)^{n-1}n$ takes the value:
- (i) $-\frac{n}{2}$ if n is even and
- (ii) $\frac{n+1}{2}$ if n is odd.

Question 4. (16 Marks) (Use writing booklet 4)

- (a) The polynomial equation $x^3 - 5x^2 + 2x - 12 = 0$ has roots α, β , and γ
Find the value of:
- (i) $\alpha + \beta + \gamma$ 1
- (ii) $\alpha\beta + \beta\gamma + \gamma\alpha$ 1
- (iii) $\alpha^2 + \beta^2 + \gamma^2$ 2
- (iv) $\alpha^3 + \beta^3 + \gamma^3$ 3
- (b) The polynomial $P(x) = 2x^3 + px + q$ has real roots α, β and $\frac{1}{\beta}$ 3
Show that $\frac{\beta^2 + 1}{\beta} = \frac{(2 - p)}{q}$
- (c) The polynomial $P(x) = mx^4 + nx^3 - 6x^2 + 22x - 12 = 0$ has a double root at $x = 1$.
- (i) Find the values of m and n 3
- (ii) Hence or otherwise factorise $P(x)$ completely. 3

End of Paper



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Solutions.

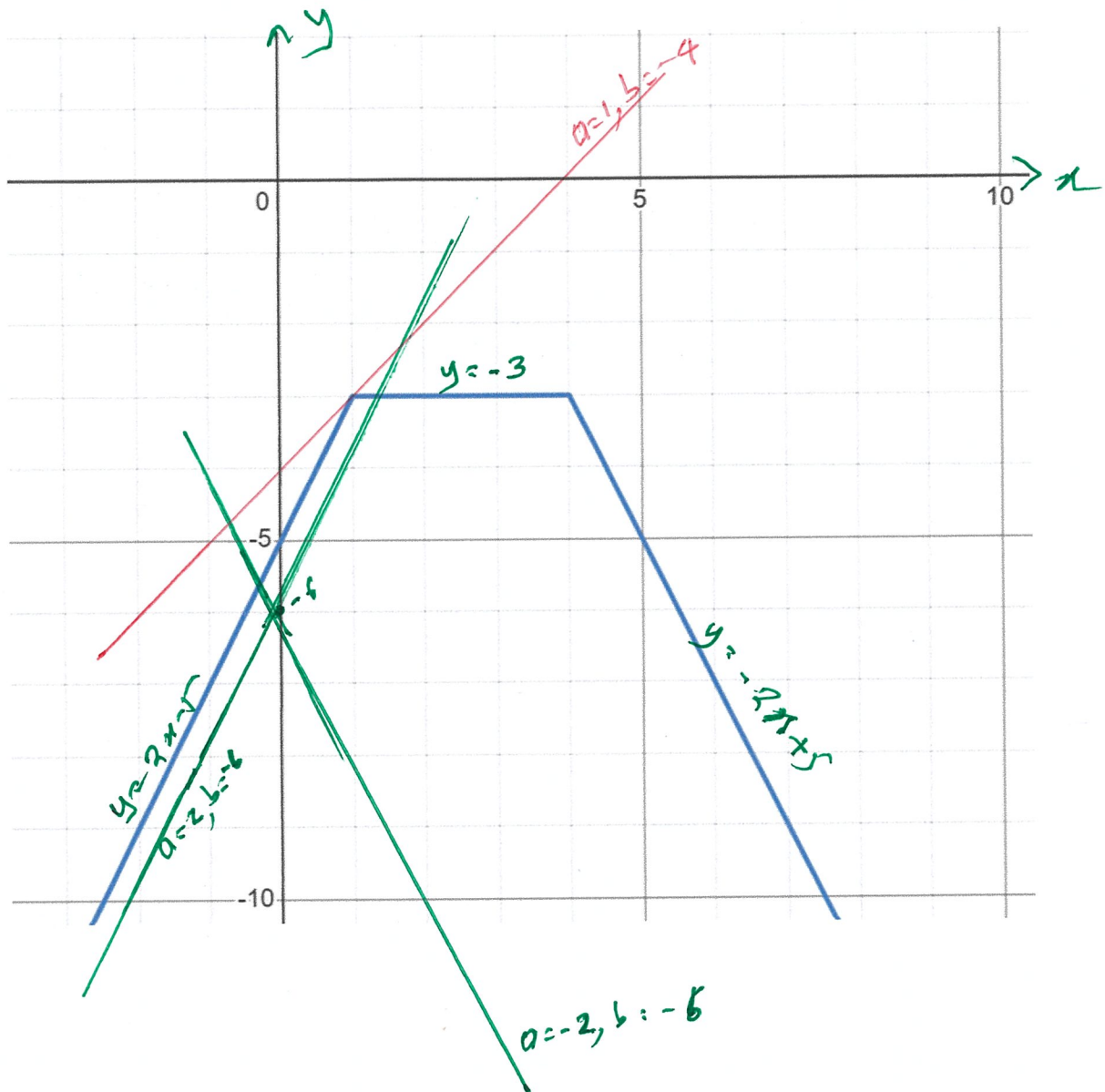
Marker's use only.

** Markers: you are at your liberty to allocate the marks for the*

QUESTION	1	2	3	4	Total	%
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Questions that you are marking,

Q, a (iv), b(i) and b(ii)



(a) (i) $A=1, B=2, C=-5$

(ii) $D=4, E=-3$

(iii) $F=-2, G=5$

(b)(i) $a=1$ and The equation *1 has one or more solutions

$$b \leq -4 \Rightarrow H = -4$$

(ii) $b = -6$ and The equation *1 has two distinct solutions

$$-2 < a < 2 \Rightarrow J = -2, L = 2$$

Q₂) (i) $\frac{6!}{2!4!} = 15$

(ii) 4 words

4W + 0B	1
3W + 1B	$\frac{4!}{3!} = 4$
2W + 2B	$\frac{4!}{2!2!} = 6$
TOTAL 11	

(iii) $\frac{9!}{5!} = 3024$

(iv) 1-digit numbers = 7
 2-digit numbers $7 \times 6 = 42$
 3-digit numbers $7 \times 6 \times 5 = 210$
 For 4-digit numbers, the 1st digit must be {1, 2 or 3}
 $\therefore$ 4-digit numbers $3 \times 6 \times 5 \times 4 = 360$
 $\therefore$ No of different numbers = $7 + 42 + 210 + 360$
 $= \underline{\underline{619}}$

(v) To be an even number, the last digit must be {2, 4 or 6}

$\therefore$ 1-digit numbers = 3
 2-digit numbers $6 \times 3 = 18$
 3-digit numbers $6 \times 5 \times 3 = 90$

For the 4-digit numbers
 If the first digit is 2, then the last digit is selected from {4 or 6} $\Rightarrow 1 \times 5 \times 4 \times 2 = 40$ numbers
 If the first digit is 1 or 3, then the last digit is selected from {2, 4 or 6} $\Rightarrow 2 \times 5 \times 4 \times 3 = 120$ numbers
 $\therefore$ No of even numbers = $3 + 18 + 90 + 40 + 120$
 $= \underline{\underline{271}}$

Q3

(a)

$$\left(2x^2 - \frac{1}{x^3}\right)^{15}$$

$$(r+1)\text{th term } T_{r+1} = \binom{15}{r} (2x^2)^{15-r} \left(-\frac{1}{x^3}\right)^r$$
$$= \binom{15}{r} 2^{15-r} (-1)^r \cdot x^{30-5r}$$

$$\text{For independent term } 30-5r = 0$$
$$\Rightarrow r = 6$$

$\therefore$ This independent term is

$$\binom{15}{6} 2^9$$
$$= \underline{\underline{2,562,560}}$$

b) ~~find~~ $(2+3x)^n$

Coef. of 2nd term : Coef. of 3rd term = 4 : 45

$$\Rightarrow \binom{n}{1} 2^{n-1} 3^1 : \binom{n}{2} 2^{n-2} 3^2 = 4 : 45$$

$$\Rightarrow \frac{n \cdot 2^{n-1} \cdot 3}{\frac{n(n-1)}{2} 2^{n-2} 3^2} = \frac{4}{45}$$

$$\Rightarrow \frac{4}{(n-1) \cdot 3} = \frac{4}{45}$$

$$\Rightarrow \frac{4}{(n-1) \cdot 3} = \frac{4}{45} \quad n-1 = 15$$
$$\underline{\underline{n = 16}}$$

Q3 (r)

$$1 + x + x^2 + \dots + x^n = \frac{1 - x^{n+1}}{1 - x}$$

Differentiating both sides w.r.t x , yield

$$1 + 2x + 3x^2 + \dots + nx^{n-1} = \frac{(1-x)[-(n+1)x^n] + (1-x^{n+1})}{(1-x)^2}$$

$$= \frac{-(n+1)x^n + nx^{n+1} + 1}{(1-x)^2}$$

When $x = -1$

$$1 - 2 + 3 - \dots + (-1)^{n-1}n = \frac{-(n+1)(-1)^n + n(-1)^{n+1} + 1}{2^2}$$

If n is even RHS = $-\frac{(n+1) - n + 1}{4}$

$$= -\frac{2n}{4}$$

$$= -\frac{n}{2}$$

If n is odd RHS = $\frac{(n+1) + n + 1}{2^2}$

$$= \frac{n+1}{2}$$

Q4

(a) $x^3 - 5x^2 + 2x - 12 = 0$ α, β, γ

i) $\alpha + \beta + \gamma = -\frac{(-5)}{1} = 5$

ii) $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{2}{1} = 2$

iii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$
 $= (5)^2 - 2(2)$
 $= \underline{\underline{21}}$

(iv) α is a root

$\therefore \alpha^3 - 5\alpha^2 + 2\alpha - 12 = 0$ — ①

ii) $\beta^3 - 5\beta^2 + 2\beta - 12 = 0$ — ②

$\gamma^3 - 5\gamma^2 + 2\gamma - 12 = 0$ — ③

① + ② + ③ $(\alpha^3 + \beta^3 + \gamma^3) - 5(\alpha^2 + \beta^2 + \gamma^2) + 2(\alpha + \beta + \gamma) - 36 = 0$
 $- 36 = 0$

$\Rightarrow (\alpha^3 + \beta^3 + \gamma^3) - 5(21) + 2(5) - 36 = 0$

$\Rightarrow \alpha^3 + \beta^3 + \gamma^3 = \underline{\underline{131}}$

(b) $p(x) = 2x^3 + px + q$ $\alpha, \beta, \frac{1}{\beta}$

Product of roots $\alpha \cdot \beta \cdot \frac{1}{\beta} = -\frac{q}{2} \Rightarrow \alpha = -\frac{q}{2}$

Product of roots 2 at a time

$\alpha\beta + \frac{\alpha}{\beta} + 1 = \frac{p}{2}$

$\alpha(\beta + \frac{1}{\beta}) = \frac{p}{2} - 1 \Rightarrow \frac{\beta^2 + 1}{\beta} = \frac{p-2}{2\alpha} = \frac{(2-q)}{2}$

4(c)

$$(i), P(x) = mx^4 + nx^3 - 6x^2 + 22x - 12 = 0$$

$P(x)$ has a double root at $x=1 \Rightarrow P(1)=0$ and $P'(1)=0$

$$P(1)=0 \Rightarrow m+n=-4 \quad \text{--- (1)}$$

$$P'(1)=0 \Rightarrow 4m+3n=-10 \quad \text{--- (2)}$$

Solving (1) and (2) simultaneously,

$$m=2, n=-6$$

$$(ii), 2x^4 - 6x^3 - 6x^2 + 22x - 12 = (x^2 - 2x + 1)(2x^2 + px + q)$$

matching the coefficients of

$$x^0: \text{ gives } q = -12$$

$$x^3: \text{ gives } p+q = -6 \Rightarrow p = -2$$

$$\therefore P(x) = (x-1)^2 (2x^2 - 6x - 12)$$

$$= 2(x-1)^2 (x-3)(x-2)$$